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Pricing of Climate Risk Insurance: Regulation and Cross-Subsidies

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	Insurance operations and insurer balance sheets	4
3.2	Rate filings and rate changes	4
3.3	Regulatory friction and rate wedge	4
3.4	Regulatory rules	5
3.5	Loss variables	5
3.6	ZIP-level pricing and border data	5
4	Models and empirical specifications	5
4.1	Rate response to same-state and out-of-state losses	5
4.2	Asymmetric spillover test	6
4.3	Instrumental variables	6
4.4	Border discontinuity design	6
4.5	Financing-friction heterogeneity	6
4.6	Long-run rate-risk decoupling and exits	6
5	Identification and empirical design	7
5.1	Core identification challenge	7
5.2	Why same-insurer comparisons matter	7
5.3	Why OSL is useful	7
5.4	Why the border design is useful	7
5.5	Mechanism tests	7
5.6	Limitations	7
6	Section-by-section study guide	8
7	Tables and figures	9
7.1	Table I: Summary Statistics and Figure 1: Distribution of Rate Wedge	9
7.2	Table II: Regulatory Pricing Rules and the Friction Measure and Figure 2: Insurers' Profitability by State Types	9
7.3	Table III: Rate Responses to Same-State and Out-of-State Losses	10
7.4	Table IV: Asymmetric Rate Spillovers and Table V: IV Analysis	11
7.5	Table VI: Border Discontinuity Design and Table VII: Financing Frictions from Border Design	12

7.6	Table VIII: Financing Frictions from Rate Filings and Table IX: Competition	12
7.7	Figure 3: Decoupling of Insurance Rates from Risk and Figure 4: Actual and Counterfactual Rate Growth	13
7.8	Table X: Exits and Cancellations	14
8	Result map	15
9	Short summary	15
10	Conclusion	16
10.1	Core conclusion	16
10.2	Economic intuition	16
10.3	Incremental contribution revisited	16
10.4	Policy implications	16
10.5	Limitations	16
10.6	Takeaway	16

1 Big picture

- **Question.** How does heterogeneous state-level rate regulation affect homeowners insurance pricing, climate-risk pass-through, cross-state risk sharing, and insurer exits?
- **Setting.** Homeowners insurance is regulated at the state level in the United States. Insurers frequently operate across many states, so the same insurer can face strict rate-setting frictions in one state and more flexible regulation in another.
- **Core mechanism.** When regulators in high-friction states restrict rate increases after losses, insurers cannot fully raise rates where risk increases. Multi-state insurers then partially compensate by raising rates in lower-friction states.
- **Main object.** The paper constructs a novel regulatory-friction measure using the gap between an insurer's targeted rate change and the rate change it actually receives in regulatory filings. A lower received/target ratio indicates stronger rate-setting friction.
- **Main empirical result.** In high-friction states, insurers are less responsive to same-state losses. In low-friction states, rates respond to out-of-state losses, especially when those losses come from high-friction states.
- **Identification strategy.** The paper combines state-insurer rate filing panels, regulatory-rule instruments, and a border discontinuity design around state borders to separate regulation-driven price distortions from local risk or insurer heterogeneity.
- **Economic mechanism.** Financing constraints matter. Constrained insurers respond more strongly to out-of-state losses, consistent with cross-subsidization being driven by capital needs after losses.
- **Long-run implications.** Rates and risk become increasingly decoupled. High-friction states have rates that are too flat relative to expected losses, while low/medium-friction states bear spillover costs. Insurers are also more likely to exit or cancel in high-friction states.
- **Economic takeaway.** Climate-risk insurance pricing is shaped not just by local risk, but by the interaction of regulation, multi-state insurer balance sheets, and financing frictions. Cross-state regulation can redistribute climate risk across households.

2 Incremental contribution of the paper

- **New market and question.** The paper studies homeowners insurance pricing, a market that is central to climate-risk adaptation and mortgage-market stability but historically less studied than health, life, or catastrophe reinsurance markets.
- **Novel regulatory-friction measure.** Instead of coding regulation only from statutes, the authors use realized rate filings to measure the gap between actuarial target rates and regulator-approved rates. This captures both formal rules and enforcement intensity.
- **Cross-jurisdiction subsidization.** The paper shows that heterogeneous product-market regulation generates cross-state subsidies through multi-state firms. Households in less regulated states partly bear the insurance costs of households in more regulated states.
- **Combines rules, filings, and border design.** The paper links hand-collected regulatory rules, insurer-state-year filings, ZIP-level pricing data, and border discontinuity evidence. This triangulation is a major empirical contribution.
- **Distinguishes same-state pass-through from spillovers.** Prior work often studies whether losses affect premiums. This paper decomposes responses into local pass-through and out-of-state spillover pass-through.

- **Mechanism evidence from financing constraints.** The paper shows that financially constrained insurers respond more strongly to out-of-state shocks, linking product prices to insurer balance sheet constraints.
- **Climate-risk pricing distortion.** The paper documents long-run decoupling between insurance rates and expected losses, showing how regulation can distort spatial climate-risk signals.
- **Availability and exit margin.** The paper moves beyond pricing by showing that high-friction states are more likely to experience exits/cancellations, which speaks to insurance availability as well as affordability.
- **Broader contribution to financial regulation.** The results generalize to markets where geographically fragmented price regulation interacts with national/multi-jurisdiction firms.

Incremental contribution in one sentence. The paper shows that state-level rate regulation of climate-exposed homeowners insurance creates a hidden cross-state subsidy system: constrained prices in high-friction states are partly financed by higher prices in low-friction states, with long-run consequences for risk signals and insurance availability.

3 Data and measurement

3.1 Insurance operations and insurer balance sheets

- The paper uses property-and-casualty insurer underwriting data by line of business and state, focusing on homeowners insurance.
- Premiums measure the total dollar amount of insurance sold; losses measure claims paid for insured events.
- Loss ratio is losses divided by premiums and proxies for underwriting cost pressure.
- Insurer financial statements supply net assets, RBC ratios, leverage, and reinsurance measures.
- These variables are used to classify insurers' financial constraints and to control for insurer balance-sheet conditions.

Interpretation: The balance sheet variables are not merely controls. They help show that cross-subsidies arise because insurers need to finance losses when regulation blocks price increases in some states.

3.2 Rate filings and rate changes

- Insurers must file rate change requests with state regulators when updating homeowners insurance rates.
- The filing data report target rate changes, which reflect insurers' actuarial assessments.
- The filing data also report received rate changes, which reflect the rates regulators approve.
- The sample covers 49 states plus Washington, D.C., mostly from 2009 to 2019.

Interpretation: The filing data are the empirical backbone of the paper. They allow the authors to observe both what insurers say they need and what regulators allow them to charge.

3.3 Regulatory friction and rate wedge

The paper defines a rate wedge as approximately:

$$RateWedge_{ist} = \frac{Rate\Delta Received_{ist}}{Rate\Delta Target_{ist}}.$$

A lower wedge means that regulators approved less of the insurer’s actuarially targeted rate increase. The paper aggregates historical filing wedges into a state-level friction measure.

Interpretation: A key insight is that regulation cannot be summarized by a single statutory category. The realized wedge captures formal rules, enforcement, political pressure, and unobserved regulatory practice.

3.4 Regulatory rules

The paper hand-collects state-level rules in four broad categories:

- restrictions on catastrophe models;
- input restrictions, including claims-history restrictions and household-characteristic restrictions;
- pricing flexibility, including ease of minor rate changes;
- enforcement pressure, including political pressure and regulator staffing.

Interpretation: The rules are used both to validate the friction measure and as instruments for regulatory friction in the IV design.

3.5 Loss variables

- Same-state losses (SSL) are lagged losses in the filing state, scaled by lagged premiums in the filing state.
- Out-of-state losses (OSL) are lagged losses in all other states where the same insurer operates, scaled by lagged premiums outside the filing state.
- OSL is split by whether losses originate in high-, medium-, or low-friction states.

Interpretation: The SSL/OSL distinction is essential. SSL measures local cost pass-through. OSL measures cross-state spillover pricing.

3.6 ZIP-level pricing and border data

- The paper uses granular ZIP-level homeowners insurance rates from insurer pricing data.
- The border discontinuity design compares neighboring ZIP codes across state borders when a nonlocal shock affects insurers that operate in the area.
- The outcome is log insurance rates at the insurer-ZIP-year level.

Interpretation: The border design is valuable because neighboring ZIPs can have similar local risk and demand but different regulatory regimes.

4 Models and empirical specifications

4.1 Rate response to same-state and out-of-state losses

The baseline filing regressions are variants of:

$$Y_{ist} = \gamma Loss_{ist-1} + \gamma^M Loss_{ist-1} \times Med_s + \gamma^L Loss_{ist-1} \times Low_s + X'_{ist} \Gamma + \alpha_{is} + \alpha_{st} + \varepsilon_{ist}.$$

Here Y_{ist} is either an indicator for any filing or the rate change received. Loss is SSL or OSL. The omitted state group is high-friction states.

Interpretation: Positive interactions with Low_s mean that low-friction states respond more to losses than high-friction states.

4.2 Asymmetric spillover test

The paper decomposes out-of-state losses by source-state regulation:

$$Y_{ist} = \beta^H OSL_{ist-1}^H + \beta^M OSL_{ist-1}^M + \beta^L OSL_{ist-1}^L + X'_{ist}\Gamma + \alpha_{is} + \alpha_{st} + \varepsilon_{ist}.$$

Interpretation: The key prediction is $\beta^H > 0$ for filings in low-friction states: low-friction states absorb losses from high-friction states.

4.3 Instrumental variables

The first stage uses regulatory rules to predict friction:

$$Friction_s = \pi_0 + Z'_s\pi + u_s,$$

where Z_s includes restrictions on CAT models, pricing inputs, pricing flexibility, and enforcement. The second stage replaces measured friction with predicted friction and estimates the loss response.

Interpretation: The IV design isolates variation in friction arising from regulatory rules rather than from endogenous insurer behavior or state shocks.

4.4 Border discontinuity design

The ZIP-level event specification is:

$$\log(Rate_{iz(e)bt}) = \beta Low_z \times Post_e \times Affected_i + controls + fixed\ effects + \varepsilon_{iz(e)bt}.$$

Here Low_z indicates a ZIP code in a low-friction state, $Post_e$ indicates after an out-of-state shock, and $Affected_i$ identifies insurers affected by the nonlocal shock.

Interpretation: The coefficient asks whether the same shock leads to higher rates on the low-friction side of a border relative to the high-friction side.

4.5 Financing-friction heterogeneity

The paper interacts out-of-state shocks with measures of financial constraints such as low RBC, high leverage, and increases in leverage.

$$Y_{ist} = \beta_1 OSL_{ist-1}^{big} \times Low_s \times Constrained_{it-1} + \dots$$

Interpretation: If cross-subsidies are driven by capital needs, financially constrained insurers should pass through out-of-state shocks more strongly.

4.6 Long-run rate-risk decoupling and exits

The paper uses cross-sectional and long-run analyses to compare rates with expected losses and to study exits/cancellations:

$$Exit_s = \alpha + \beta HighFriction_s + X'_s\Gamma + \varepsilon_s.$$

Interpretation: These tests examine consequences, not just mechanisms: regulation can distort risk prices and affect insurance availability.

5 Identification and empirical design

5.1 Core identification challenge

The key challenge is that high- and low-friction states may differ in risk, insurer mix, consumer demand, or market power. The paper addresses this by comparing the same insurer across states, using insurer-state fixed effects, state-year fixed effects, rule-based instruments, and ZIP-border comparisons.

5.2 Why same-insurer comparisons matter

Because a multi-state insurer files rates in many states, the paper can compare how the same insurer responds to losses under different regulatory regimes. This reduces confounding from insurer-level risk tolerance, business model, and actuarial technology.

5.3 Why OSL is useful

Out-of-state losses are not local cost shocks for the filing state. If filing-state rates rise after OSL, the response cannot simply be standard local loss pass-through. It points to cross-subsidization or insurer-level financing needs.

5.4 Why the border design is useful

ZIPs near a border can have similar geography, housing markets, and local risk but different state regulation. The design asks whether insurers affected by nonlocal shocks raise rates more on the low-friction side of the border.

5.5 Mechanism tests

The paper rules out several alternatives:

- the results are not solely driven by competition, because spillovers appear in different competition groups;
- product quality changes are not the main explanation, because contract feature changes do not move similarly;
- operation-size differences do not drive results, because several subsample and control exercises show similar effects;
- financing constraints matter, as constrained insurers respond more strongly to out-of-state shocks.

5.6 Limitations

The paper does not estimate full welfare or optimal regulation. It documents pricing distortions and cross-subsidies, but does not claim that rate regulation is always undesirable. Regulation may address insurer market power or affordability concerns. The paper's point is that heterogeneous state regulation has cross-state consequences.

6 Section-by-section study guide

- **Introduction.** Provides the central cross-subsidy thesis and motivates why homeowners insurance pricing is important for climate-risk adaptation.
- **Institutional details.** Explains state-level rate regulation and why regulatory rules differ across states.
- **Data and measurement.** Introduces underwriting data, rate filings, target and received rates, rate wedge, friction index, and regulatory rules.
- **Main filing regressions.** Shows that high-friction states pass through local losses less and that low-friction states respond to out-of-state losses.
- **IV analysis.** Uses regulatory rules as instruments for the friction measure.
- **Border discontinuity design.** Uses ZIP-level rate data around borders to show within-local-market spillovers.
- **Mechanism tests.** Shows the role of financing constraints and rules out competition, product-quality changes, and state-size explanations.
- **Long-run implications.** Documents rate-risk decoupling, counterfactual rate indices, and insurer exits/cancellations.

7 Tables and figures

7.1 Table I: Summary Statistics and Figure 1: Distribution of Rate Wedge

Read:

- Table I summarizes underwriting operations, financial statements, and rate filings.
- Average homeowners loss ratio is around 0.57 in 2019.
- In rate filings, the mean target rate change is much larger than the mean received rate change.
- Figure 1 shows many filings with rate wedge below one, meaning regulators often approve less than insurers target.
- There is also a spike near one, showing that some requested changes are fully approved.

Detailed interpretation: Table I and Figure 1 introduce the paper’s central measurement idea. The target rate captures the insurer’s actuarial assessment; the received rate incorporates regulatory approval. The gap between them becomes a revealed-preference measure of regulation. This is more informative than a single rule because it captures how rules are implemented in practice.

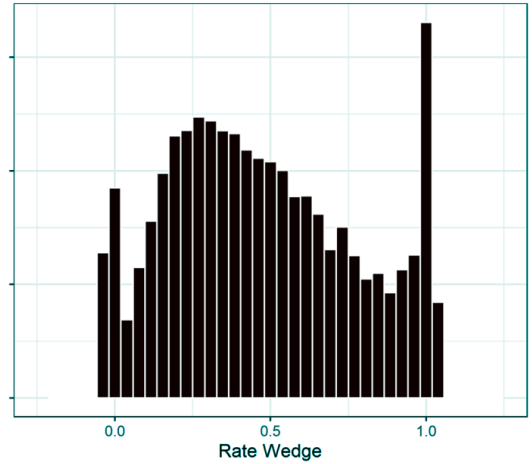
Economic message: Regulatory frictions are visible in actual pricing outcomes: insurers frequently receive rate changes below their actuarial targets.

Table I
Summary Statistics

The table presents summary statistics. Variable descriptions are as follows. *Underwriting operations:* Premiums refer to the total amount of insurance sold. Losses refer to the total claims insurers pay to consumers if an insured event takes place. Loss ratio is the ratio of losses to premiums sold. These data are at the insurer $\times$ state $\times$ year level. We report statistics for the year 2019. *Financial statements:* We report loss ratio for nonhomeowners’ lines. Net assets is total assets of the insurer. RBC ratio is the ratio of total available capital to total required capital. Reinsurance ratio is the fraction of premiums reinsured. N states is the number of states insurers sell HO insurance in. These data are at the insurer $\times$ year level. We report statistics for the year 2019. *Rate filings:* Any Filings refers to whether an insurer filed for a rate change in a given state and year. *Rate Δ Target* and *Rate Δ Received* are the rate change targeted and received by an insurer in a given state and year. These two variables are populated only conditional on filing. These data are at the insurer $\times$ state $\times$ year level. We report statistics on the full sample from 2009 to 2019. The statistics pertain to our main sample as detailed in Section B of the [Internet Appendix](#).

	N	Mean	SD	10 th	25 th	50 th	75 th	90 th
Underwriting operations (2019):								
Premium (HO, \$M)	1,530	39.03	95.72	1.90	4.95	13.01	33.53	79.77
Loss (HO, \$M)	1,530	22.39	62.11	0.80	2.54	6.55	17.62	46.87
Loss ratio (HO)	1,530	0.57	0.31	0.30	0.40	0.52	0.66	0.86
Financial statements (2019):								
Loss ratio (nonhomeowners)	253	0.60	0.15	0.49	0.54	0.58	0.64	0.71
Net assets (\$B)	253	2.86	7.33	0.07	0.13	0.33	1.75	6.79
RBC ratio (logged)	253	6.58	0.55	5.92	6.18	6.57	6.91	7.24
Reinsurance ratio	253	0.18	0.25	0.00	0.01	0.08	0.22	0.51
N states a firm sells HO	253	14.62	16.04	2.00	3.00	7.00	21.00	45.80
Rate filings (2009–2019):								
Any Filings	17,980	0.70	0.46	0.00	0.00	1.00	1.00	1.00
Rate Δ Target (%)	12,563	15.60	15.97	0.00	5.20	11.70	21.44	35.59
Rate Δ Received (%)	12,563	5.85	5.55	0.00	2.03	5.00	8.70	12.86

(a) Table I: Summary Statistics.



(b) Figure 1: Distribution of rate wedge.

7.2 Table II: Regulatory Pricing Rules and the Friction Measure and Figure 2: Insurers’ Profitability by State Types

Read:

- Table II relates the friction measure to hand-collected regulatory rules.
- CAT model restrictions, claims-history restrictions, pricing flexibility, and political pressure are positively related to friction.

- The multivariate specification has high explanatory power, confirming that formal rules explain much of the friction measure.
- Figure 2 shows underwriting profitability is lowest in high-friction states and highest in low-friction states.

Detailed interpretation: Table II validates the friction measure: it is not just noise in filing data, but is systematically related to regulatory restrictions. Figure 2 shows that those restrictions bind economically, not only administratively. High-friction states are less profitable for insurers, consistent with rate constraints limiting pass-through.

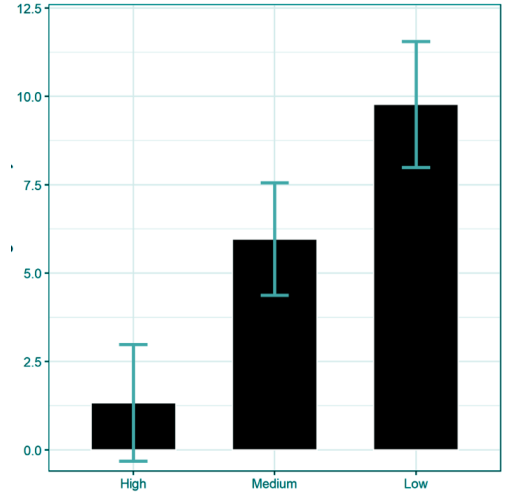
Economic message: The friction measure captures real pricing constraints. Regulatory rules translate into weaker underwriting profitability.

Table II
Regulatory Pricing Rules and the Friction Measure

The table shows the relationship between the friction measure ($Friction_{it}$) and the regulatory pricing rules. Column (1) shows the coefficients from running individual univariate regressions between $Friction_{it}$ and each regulatory rule, $Friction_{it} = \alpha + \gamma Z_{it} + \epsilon_{it}$. Column (2) shows the corresponding R^2 of each univariate regression. Column (3) shows the results of running a multivariate regression using all of the regulatory rules as regressors. R^2 and F -statistic for the joint significance of the model are shown in the bottom panel. Standard errors are reported in parentheses. A detailed description of the regulatory rules can be found in Table IA.VI. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Variable	Type	Univariate		Multivariate Coefficient (3)
		Coefficient (1)	Adjusted R^2 (2)	
CAT model restriction	Model	0.047*** (0.013)	0.201	0.024** (0.010)
Claims history restriction	Inputs	0.072*** (0.013)	0.377	0.055*** (0.011)
No. of restricted household chars	Inputs	0.021*** (0.007)	0.153	0.007 (0.005)
Territorial restriction	Inputs	0.022** (0.009)	0.086	0.008 (0.008)
Pricing flexibility	Flexibility	0.072** (0.032)	0.079	0.038 (0.027)
Political pressure	Enforcement	0.052** (0.021)	0.100	0.039** (0.015)
Observations		50		50
Adjusted R^2				0.596
F -statistic				13.03

(a) Table II: Regulatory Pricing Rules and the Friction Measure.



(b) Figure 2: Profitability by state type.

7.3 Table III: Rate Responses to Same-State and Out-of-State Losses

Read:

- Same-state losses lead to stronger filing and rate responses in low-friction states than in high-friction states.
- Out-of-state losses also affect rates, especially in low-friction states.
- The low-friction interaction for out-of-state losses is positive and economically large.
- The results include insurer-state and state-year fixed effects.

Detailed interpretation: Table III is the first main result table. The same insurer responds differently depending on the regulatory environment of the filing state. High-friction states limit local pass-through. Low-friction states become the place where losses from the insurer's overall portfolio can be recovered.

Economic message: Rates are not purely local prices for local risk. Multi-state insurers use flexible states to absorb losses from elsewhere.

Table III
Rate Responses to Same-State and Out-of-State Losses

The table presents the results from estimating equation (3) in columns (1) and (2) and equation (4) in columns (3) and (4). The dependent variable in odd columns is whether a rate change is filed and in even columns is the rate change received by insurer i in state s and year t . The independent variable in columns (1) and (2) is same-state losses (SSL), that is, losses experienced by an insurer in the state in which it has made the rate filing. SSLs are lagged one year and scaled by the lagged total premium sold in the filing state. The independent variable in columns (3) and (4) is out-of-state losses (OSL). To compute OSL, we sum an insurer's lagged losses in all the states it operates in other than the filing state s , which we scale by lagged total premiums sold in all states except s . The indicator variables Med_s and Low_s equal one if the filing state s is, correspondingly, a medium or a low friction state, respectively. All regressions control for log assets, log RBC ratio, loss ratio of all other (nonhomeowners') lines of business, percent of premiums reinsured for insurer i in year t , and interactions of the loss measure (SSL or OSL) with state size and with state risk. In addition, column (2) also controls for the insurer's target rate change, and columns (3) and (4) control for same-state losses. The results pertain to our main sample as detailed in Section B of the Internet Appendix. All regressions include insurer-filing state and filing state-year of submission fixed effects. Standard errors are shown in parentheses, clustered at the state level. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

	Same-State Losses (SSL)		Out-of-State Losses (OSL)	
	Any Filings _{ist} (1)	Rate Δ Received _{ist} (2)	Any Filings _{ist} (3)	Rate Δ Received _{ist} (4)
$Loss_{ist-1}$	0.027 (0.061)	0.139 (0.442)	0.047 (0.058)	-0.108 (0.726)
$Loss_{ist-1} \times Med_s$	0.063* (0.036)	0.030 (0.408)	0.020 (0.026)	0.651* (0.341)
$Loss_{ist-1} \times Low_s$	0.106** (0.042)	0.755** (0.310)	0.150*** (0.036)	1.568*** (0.532)
E[LHS]	0.7	3.7	0.7	3.7
State type	All	All	All	All
Controls	Yes	Yes	Yes	Yes
State $\times$ Year Fixed effects	Yes	Yes	Yes	Yes
Insurer $\times$ State Fixed effects	Yes	Yes	Yes	Yes
Observations	17,980	17,980	17,980	17,980

Table III: Rate Responses to Same-State and Out-of-State Losses.

7.4 Table IV: Asymmetric Rate Spillovers and Table V: IV Analysis

Read:

- Table IV decomposes out-of-state losses by whether they come from high-, medium-, or low-friction states.
- Low-friction states respond to losses from high-friction states; the reverse is not found.
- Table V instruments regulatory friction with rule-based predicted friction.
- The IV estimates show that higher friction reduces the responsiveness to same-state and out-of-state losses.

Detailed interpretation: These tables show the cross-subsidy direction. The spillover is not generic across all states. It is specifically high-to-low: losses in high-friction states are priced into low-friction states. The IV exercise further strengthens the interpretation by isolating rule-driven variation in friction.

Economic message: Regulatory heterogeneity creates asymmetric redistribution: low-friction households partly finance the risks of high-friction households.

Table IV
Asymmetric Rate Spillovers: Splitting Out-of-State Losses by State Type

The table presents results from estimating equation (5). The dependent variables are: in column (1), whether a rate change is filed, and in column (2) the rate change received by insurer i in the filing state s and year t . The main independent variable is insurer's lagged out-of-state losses (OSL), which we split into losses coming from high (OSL^H), medium (OSL^M), low (OSL^L) friction states, scaled by total lagged out-of-state premiums from all states. The results pertain to our main sample as detailed in Section B of the Internet Appendix. All regressions include fixed effects and controls as in Table III. Standard errors are shown in parentheses, clustered at the state level. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

	Any Filings _{ist} (1)	RateΔReceived _{ist} (2)
OSL^H_{ist-1}	0.485*** (0.098)	3.843** (1.401)
OSL^M_{ist-1}	0.407** (0.146)	1.928 (1.518)
OSL^L_{ist-1}	-0.040 (0.382)	1.048 (5.261)
E[LHS]	0.7	3.3
State type	Low	Low
Controls	Yes	Yes
State × Year Fixed effects	Yes	Yes
Insurer × State Fixed effects	Yes	Yes
Observations	6,093	6,093

(a) Table IV: Asymmetric Rate Spillovers.

Table V
Rate Responses to Same-State and Out-of-State Losses: IV Analysis

The table presents results of estimating the instrumental variable regression described in equation (7). $\widehat{Friction}_s$ is the predicted friction from equation (6). The variable $Loss$ refers to same-state losses (SSL) (columns (1) and (2)) and out-of-state losses (OSL) (columns (3) and (4)). SSL is lagged one year and scaled by the lagged total premium in the filing state. To compute OSL, we sum an insurer's lagged losses in all states it operates in other than the filing state s , which we scale by lagged total premiums sold in all states except s . The dependent variables are a binary indicator for rate change filing (odd columns) or the magnitude of the rate change (even columns). Control variables are the same as those included in the main specifications. Standard errors are obtained through bootstrapping to account for the estimation error in the first stage. Fixed effects are shown at the bottom. The sample is as detailed in Section B of the Internet Appendix, excluding single-state insurers. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

	Same-State Losses (SSL)		Out-of-State Losses (OSL)	
	Any Filings _{ist} (1)	RateΔReceived _{ist} (2)	Any Filings _{ist} (3)	RateΔReceived _{ist} (4)
$Loss_{ist-1}$	0.544*** (0.059)	3.710*** (0.433)	0.283*** (0.048)	5.997*** (1.023)
$Loss_{ist-1} \times \widehat{Friction}_s$	-0.793*** (0.108)	-5.286*** (0.817)	-0.242*** (0.081)	-8.871*** (1.799)
State type	All	All	All	All
Controls	Yes	Yes	Yes	Yes
State × Year Fixed Effects	Yes	Yes	Yes	Yes
Insurer × State Fixed Effects	Yes	Yes	Yes	Yes
Observations	17,980	17,980	17,980	17,980

(b) Table V: IV Analysis.

7.5 Table VI: Border Discontinuity Design and Table VII: Financing Frictions from Border Design

Read:

- Table VI shows that low-friction ZIP codes experience stronger rate increases after nonlocal out-of-state shocks affecting insurers.
- The key triple interaction is positive across the main specifications.
- Table VII shows the effect is stronger when insurers are financially constrained.
- Financial constraints are proxied by RBC, leverage, and changes in leverage.

Detailed interpretation: The border design is powerful because neighboring ZIP codes can be similar in local risk, housing markets, and demand, but fall under different state regulation. The result therefore points to regulation rather than local risk. Table VII adds mechanism: constrained insurers use low-friction states more aggressively after shocks.

Economic message: Cross-subsidies are visible within local border markets and are amplified by insurer balance-sheet constraints.

7.6 Table VIII: Financing Frictions from Rate Filings and Table IX: Competition

Read:

- Table VIII shows that constrained insurers respond more strongly to large out-of-state shocks in low-friction states.
- The triple interactions with low friction and financial constraints are positive in several specifications.
- Table IX splits low-friction states by competition measures.
- Spillovers persist across competition subsamples, weakening a pure market-power explanation.

Table VI
Border Discontinuity Design Response to Out-of-State Shocks

This table presents results from estimating equation (8). The dependent variable is insurer rates (in logs) charged by insurer i in year t in code z , where the ZIP code belongs to a given state border pair b . Low_{it} is an indicator variable that takes the value of one for ZIP codes that are in a low friction state and zero if in a high friction state. $Post_{it}$ is an indicator variable that takes the value of one after the shock for a given shock and zero before. $Affected_{it}$ ($= 1$) identifies insurers affected by the rebordering out-of-state event c . Columns (1) to (5) report results for the n specification and columns (6) and (7) report results of robustness tests 1 and 2, as discussed in the main text. The controls are as discussed in main text. Fixed effects are denoted at the bottom of the table. Standard errors are shown in parentheses, clustered at the ZIP code level. * $p < 0.05$; ** $p < 0.01$.

	log(Insurance Rate) _{it(c)bz}						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$Low_{it} \times Post_{it} \times Affected_{it}$	0.028*** (0.005)	0.037*** (0.006)	0.064*** (0.006)	0.055*** (0.006)	0.037*** (0.006)	0.062*** (0.008)	0.065*** (0.006)
$Low_{it} \times Affected_{it}$	-0.049*** (0.011)	-0.027*** (0.007)	-0.051*** (0.008)	-0.044*** (0.008)			
$Post_{it} \times Affected_{it}$	0.024*** (0.005)	-0.051*** (0.005)	-0.072*** (0.006)				
$Low_{it} \times Post_{it}$	0.0001 (0.004)	-0.020*** (0.003)					
$Post_{it}$	0.015*** (0.004)	0.008*** (0.002)					
Low_{it}	0.140*** (0.028)	0.118*** (0.018)					
$Affected_{it}$	0.169*** (0.009)						
Specification	Main	Main	Main	Main	Main	Robustness 1	Robustness 2
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Border $\times$ Insurer $\times$ Event Fixed effects	No	Yes	Yes	Yes	No	No	No
Border $\times$ Year $\times$ Event Fixed effects	No	No	Yes	No	No	No	No
Insurer $\times$ Year $\times$ Event Fixed effects	No	No	No	Yes	Yes	Yes	Yes
ZIP $\times$ Insurer $\times$ Event Fixed effects	No	No	No	No	Yes	Yes	Yes
ZIP $\times$ Year $\times$ Event Fixed effects	No	No	No	No	Yes	Yes	Yes
Observations	290,158	290,158	290,158	290,158	290,158	258,445	155,23

Table VII
The Role of Financing Frictions: Cross-Sectional Evidence from Border Design

shows the heterogeneity in response to out-of-state shocks by reestimating equation (8) for different subset of insurers. Different proxies of financial constraints (RBC ratio, leverage, and changes in leverage). Using each proxy, we then split the groups (as of the year of the shock), above and below median, to distinguish constrained from unconstrained insurers. For each (low) leverage are defined as constrained (unconstrained). For ease of comparison, we do not split the unaffected (control) group (unaffected) remains the same. All main variables are as defined as before. All regressions include fixed effect (5) of Table VI. Standard errors are shown in parentheses, clustered at the ZIP code level. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

	log(Insurance Rate) _{it(c)bz}				
	(1)	(2)	(3)	(4)	(5)
$Post_{it} \times Affected_{it}$	0.0003 (0.005)	0.082*** (0.016)	0.019*** (0.004)	0.054*** (0.011)	0.018** (0.007)
Unconstrained	No	Yes	No	Yes	No
Constrained	RBC	RBC	Leverage	Leverage	Δ Leverage
Year $\times$ Event Fixed effects	Yes	Yes	Yes	Yes	Yes
Insurer $\times$ Event Fixed effects	Yes	Yes	Yes	Yes	Yes
Year $\times$ Event Fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	259,553	250,804	255,328	255,029	258,871

Detailed interpretation: Table VIII links rate-filing evidence to the financing-constraint mechanism. If an insurer faces capital pressure after losses, it needs to raise rates somewhere. Table IX addresses whether cross-subsidies are just a result of weak local competition. The spillover pattern remains even when low-friction states are split by competition.

Economic message: The spillover mechanism is more consistent with insurer financing constraints than with simple market power.

Table VIII

The Role of Financing Frictions: Cross-Sectional Evidence from Rate Filings

The table presents results from estimating the regression

$$Y_{ist} = (\beta + \beta^M \times Med_s + \beta^L \times Low_s) \times OSL_{ist-1}^{big} + (\beta_C + \beta_C^M \times Med_s + \beta_C^L \times Low_s) \times OSL_{ist-1}^{big} \times Constr_{it-1} + OSL_{ist-1}^{asm} + \theta X_{it} + \alpha_{is} + \alpha_{st} + \epsilon_{ist}.$$

Dependent variables are denoted at the top of the table. The independent variables are as follows. OSL_{ist-1}^{big} equals OSL if OSL is above the 90th percentile of OSL, and zero otherwise. OSL_{ist-1}^{asm} equals OSL if OSL is below the 90th percentile of OSL, and zero otherwise. OSL is an insurer's out-of-state losses, constructed as described in Table III. The indicator variables Med_s and Low_s equal zero if the state s is, correspondingly, a medium or a low friction state. $Constr_{it-1}$ equals one if insurer i is financially constrained according to the various metrics defined in Section IV.B, which are denoted at the bottom panel (RBC, leverage, and changes in leverage). All regressions include fixed effects and controls as in Table III. Standard errors are shown in parentheses, clustered at the state level. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

	Any Filings _{ist}			Rate Δ Received _{ist}		
	(1)	(2)	(3)	(4)	(5)	(6)
$OSL_{ist-1}^{big} \times Low_s \times Constr_{it-1}$	0.110** (0.048)	0.111*** (0.035)	-0.001 (0.041)	1.168** (0.471)	0.835** (0.414)	0.363 (0.488)
$OSL_{ist-1}^{big} \times Low_s$	-0.028 (0.044)	-0.023 (0.034)	0.050 (0.044)	-0.186 (0.494)	0.070 (0.443)	0.393 (0.414)
$OSL_{ist-1}^{big} \times Med_s \times Constr_{it-1}$	-0.024 (0.036)	0.009 (0.034)	0.001 (0.039)	0.359 (0.408)	0.099 (0.512)	0.523 (0.447)
$OSL_{ist-1}^{big} \times Med_s$	0.027 (0.037)	0.004 (0.039)	0.012 (0.041)	0.223 (0.407)	0.359 (0.418)	0.137 (0.417)
$OSL_{ist-1}^{big} \times Constr_{it-1}$	0.013 (0.024)	-0.033 (0.023)	-0.005 (0.031)	-0.297 (0.295)	-0.124 (0.315)	-0.182 (0.320)
OSL_{ist-1}^{big}	0.086 (0.070)	0.116 (0.070)	0.106 (0.074)	0.593 (0.872)	0.431 (0.827)	0.588 (0.877)
Firm Constraint	RBC	Leverage	Δ Leverage	RBC	Leverage	Δ Leverage
Controls	Yes	Yes	Yes	Yes	Yes	Yes
State $\times$ Year Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Insurer $\times$ State Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	17,980	17,980	17,980	17,980	17,980	17,980

(a) Table VIII: Financing Frictions from Rate Filings.

Table IX
The Role of Competition

The table presents results from estimating equation (4), where we split the low friction states into two groups, above median (H) and below median (L), by the level of competition in each market, estimated using either the fraction of the premium sold by single-state insurers (1*), HHI index (H) or the fraction of premiums sold by the largest five insurers (C5). As high and medium friction states are not split further, the control group remains the same as prior tables, that is, all high and medium friction states. Dependent variables are denoted at the top of the table. The independent variable of interest is an insurer's "out-of-state" losses (OSL) in the prior year. To compute OSL, we sum an insurer's lagged losses in all the states operated in other than the filing state s , which we scale by lagged total premiums sold in all states except s . The indicator variables Med_s and L_s equal one if the state s is, correspondingly, a medium or a low friction state. The results pertain to our main sample as detailed in Section B of Internet Appendix. All regressions include fixed effects and controls as in Table III. Standard errors are shown in parentheses, clustered at the state level. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

	Any Filings _{ist}						Rate Δ Received _{ist}					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
OSL_{ist-1}	-0.009 (0.044)	-0.021 (0.053)	0.020 (0.047)	0.034 (0.061)	0.015 (0.045)	0.031 (0.068)	-1.221* (0.610)	0.034 (0.619)	-0.784 (0.739)	-0.283 (0.560)	-0.0 (0.733)	-0.0 (0.733)
$OSL_{ist-1} \times Med_s$	0.018 (0.028)	0.018 (0.026)	0.019 (0.026)	0.019 (0.026)	0.019 (0.026)	0.019 (0.026)	0.632* (0.348)	0.640* (0.348)	0.643* (0.344)	0.639* (0.346)	0.6 (0.342)	0.6 (0.342)
$OSL_{ist-1} \times Low_s$	0.091** (0.036)	0.258*** (0.051)	0.109*** (0.037)	0.224*** (0.053)	0.146*** (0.033)	0.172** (0.066)	0.929* (0.465)	2.888*** (0.846)	1.085* (0.617)	2.247*** (0.703)	1.553** (0.588)	1.6 (0.588)
Competition Measure	H	L	H	L	H	L	H	L	H	L	H	L
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State $\times$ Year Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Insurer $\times$ State Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	14,620	15,247	15,174	14,693	15,256	14,511	14,620	15,247	15,174	14,693	15,256	14,511

(b) Table IX: Role of Competition.

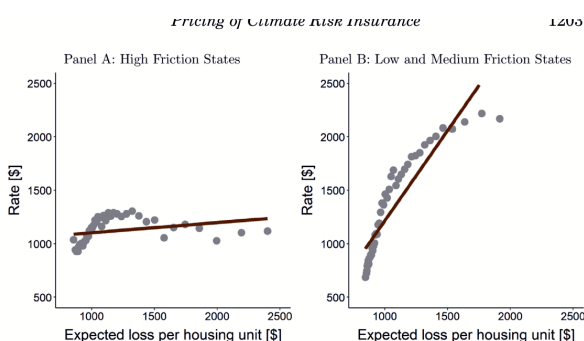
7.7 Figure 3: Decoupling of Insurance Rates from Risk and Figure 4: Actual and Counterfactual Rate Growth

Read:

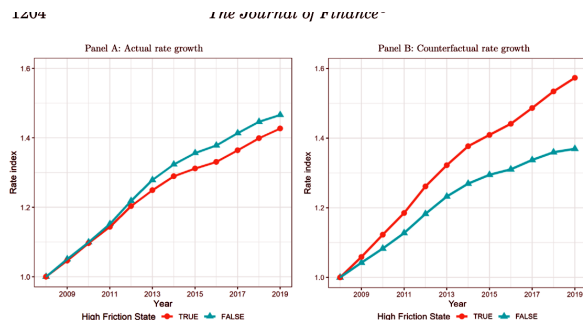
- Figure 3 shows a weak relation between rates and expected losses in high-friction states.
- In low and medium-friction states, rates are more tightly related to expected losses.
- Figure 4 shows actual rate growth differs from counterfactual scenarios that remove regulation and spillover effects.
- The counterfactuals quantify how much of long-run divergence comes from constrained high-friction prices and spillover pricing in low/medium states.

Detailed interpretation: These figures show the long-run consequence of the short-run pass-through results. If high-risk/high-friction states cannot raise rates enough, rates become too flat relative to expected losses. If low-friction states absorb spillovers, their rates rise relative to their own risk. The two forces jointly decouple insurance rates from risk.

Economic message: Regulation reshapes climate-risk price signals. Households may face premiums that do not reflect local expected losses.



(a) Figure 3: Rates versus expected losses.



(b) Figure 4: Actual and counterfactual rate growth.

7.8 Table X: Exits and Cancellations

Read:

- Table X studies hard exits and policy cancellations at the state-insurer level.
- High-friction states have higher exit/cancellation likelihoods in the aggregate evidence.
- Cross-sectional tests examine ability to cross-subsidize, regulatory pressure, bundling, and geographic diversification.
- The results suggest that exits historically occur but are moderated by bundling, pressure, and diversification.

Detailed interpretation: Table X connects price regulation to insurance availability. The paper argues that exits were historically limited in magnitude, but high-friction states face greater exit pressure. The heterogeneity tests explain why hard exits may be muted even when pricing is constrained: insurers can bundle products, diversify geographically, and face regulatory pressure not to cancel.

Economic message: Rate regulation can preserve affordability in the short run but can threaten private-market availability when climate losses rise.

Table X
Exits and Cancellations

The table presents results from estimating the regression $Y_{it} = \beta High_{it} + X_{it} + \alpha_i + \epsilon_{it}$. $High_{it}$ equals one for high friction states and zero for low friction states. The dependent variable in columns (1) and (3) to (10) is the percent of insurers who exit state s in year t . Because exits are rare, we collapse into a state $\times$ year panel. In measuring exits, we require that an insurer exits a particular state but continues to operate in at least one other state. We exclude insurers who have <0.05% market share as they tend to switch in and out of a state. The years of observation are 2009 to 2014. Column (2) is the fraction of existing policies cancelled or not renewed from NAIC's Market Conduct Annual survey at the state level from 2014. Column (1) shows all insurers; (3) shows large insurers (market share above 1%); (4) shows small insurer insurers with high reliance on the state (the state accounts for over 9% of premiums for the insurer, which is the 75th percentile of the distribution of auto and HO) for the state is between 33% and 66%; (5) shows insurers less likely to bundle (specialize in either auto or HO ins and (10) show insurers whose HO business is above and below the median in geographic diversification of the HO line, respectively, measured by HHI shares. Controls include states' median household income, percentage of population that is black or Hispanic, percentage of of the state from insurance, average percentage of of republican votes, total HO insurers' net assets, and average RBC ratio of HO insurers. Standard errors shown in parentheses, clustered at the state level. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

	Aggregate Evidence		Cross-Sectional Evidence on Exits						
	Exits (All) (1)	Cancellations (2)	Ability to Cross-Sub		Regulatory Pressure		Bundling		Divergence (9)
			Large (3)	Small (4)	More (5)	Less (6)	More (7)	Less (8)	
High	0.139*** (0.021)	0.225*** (0.073)	0.063 (0.072)	0.240** (0.096)	0.059 (0.066)	0.095*** (0.030)	-0.074 (0.068)	0.224*** (0.035)	0.057 (0.145)
E[LHS]	0.28	3.29	0.18	0.45	0.28	0.21	0.26	0.29	0.24
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	500	284	500	499	500	500	500	500	500

Table X: Exits and Cancellations.

8 Result map

- **Measurement result.** A filing-based rate wedge captures regulatory friction and is validated by hand-collected pricing rules.
- **Profitability result.** High-friction states have lower underwriting profitability.
- **Same-state pass-through result.** Insurers adjust less to local losses in high-friction states.
- **Out-of-state spillover result.** Low-friction states respond to out-of-state losses, especially losses from high-friction states.
- **IV result.** Rule-predicted regulatory friction produces the same qualitative findings.
- **Border result.** Neighboring ZIP-code comparisons show that low-friction sides of borders experience higher rates after nonlocal shocks.
- **Financing mechanism.** Financially constrained insurers pass through out-of-state losses more strongly.
- **Competition check.** The results are not explained solely by weak competition.
- **Long-run pricing implication.** Rates become decoupled from expected losses.
- **Availability implication.** High-friction states face greater exit/cancellation pressure.

9 Short summary

- The paper studies homeowners insurance rate regulation and climate-risk pricing.
- It constructs a new regulatory-friction measure from target and received rate changes.
- High-friction states have lower underwriting profitability.
- Insurers pass through same-state losses less in high-friction states.
- Insurers raise rates in low-friction states after out-of-state losses from high-friction states.

- Rule-based IV estimates and ZIP-border designs support the interpretation that regulation drives the patterns.
- Financing constraints explain why insurers cross-subsidize across states.
- The long-run result is a decoupling between rates and expected losses.
- High-friction states face increased exit and cancellation risk.
- The paper shows that local climate-risk insurance prices can be shaped by distant losses and state-level regulation.

10 Conclusion

10.1 Core conclusion

Oh, Sen, and Tenekedjieva show that state-level homeowners insurance regulation creates persistent pricing distortions and cross-state subsidies. High-friction states constrain rate increases after losses. Multi-state insurers partly compensate by raising rates in low-friction states.

10.2 Economic intuition

A multi-state insurer has one balance sheet but faces many regulators. If a high-friction regulator blocks a rate increase after losses, the insurer still needs to restore profitability and capital. Low-friction states provide the adjustment margin. Thus, regulation in one state changes prices in another state.

10.3 Incremental contribution revisited

The paper’s incremental contribution is to combine a new filing-based friction measure, hand-collected regulatory rules, cross-state insurer panels, rule-based IV, and border discontinuity evidence to show that climate-risk insurance regulation creates cross-jurisdiction subsidies and long-run rate-risk decoupling.

10.4 Policy implications

The results highlight a regulatory trade-off. Strict rate regulation can preserve affordability for households in high-risk states, but it may distort risk signals, shift costs to other states, and reduce insurer availability over time. Low-friction households may pay rates partly reflecting climate losses elsewhere.

10.5 Limitations

The paper does not estimate a complete welfare optimum. It does not claim that all rate regulation is bad; regulation may address affordability, market power, or other failures. The empirical point is that heterogeneous regulation has spillovers when insurers operate across states.

10.6 Takeaway

The central takeaway is that homeowners insurance prices do not simply reflect local risk. They reflect local risk, state regulation, insurer financing constraints, and losses across the insurer’s national portfolio. In a climate-risk economy, these interactions determine who actually bears disaster risk.